



## Mock Questions 2025/26

**MODULE CODE: COMP202**  
**MODULE TITLE: Complexity of Algorithms**

**TIME ALLOWED: TWO Hours**

|                             | Additional Information                |
|-----------------------------|---------------------------------------|
| Calculators Allowed?        | No                                    |
| Student permitted materials | Writing materials                     |
| Provided materials          | Question paper<br>1x MCQ answer sheet |

Instructions for examination:

- Prepare your answers in the question paper, selecting one option for each question
- When ready, transfer your answers to the MCQ answer sheet provided, shading in the appropriate box for each question

At the end of the examination:

- Please keep the MCQ answer sheet separate from any answer booklets
- **Leave all exam materials, including this question paper, in the examination room**

1. Which of the following is true.

**Note:**  $\log n = \log(n) = \log_2(n)$ .

- ☒ **A.** For all constants  $c > 0$ ,  $\log n = O(n^c)$ .
- ☐ **B.**  $\log n = \Omega(n^{0.5})$
- ☐ **C.** There exists a constant  $c > 0$  such that  $\log n = \Theta(n^c)$
- ☐ **D.**  $n^3 = O(n^2)$
- ☐ **E.**  $2^n = O(\log n)$

2. The function  $T(n)$  is assumed to have the initial value

$T(n) = d$ , for any  $n \leq c$ , where  $c, d \geq 1$  are some fixed integer constants.

**Note:**  $\log n = \log(n) = \log_2(n)$ .

What is the correct solution to the following recurrence relation that holds for all  $n > c$ :

$$T(n) = 4T(n/4) + 8n.$$

- ☐ **A.**  $T(n)$  is  $\Theta(n^4)$
- ☐ **B.**  $T(n)$  is  $\Theta(n^8)$
- ☐ **C.**  $T(n)$  is  $\Theta(n^2)$
- ☒ **D.**  $T(n)$  is  $\Theta(n \log n)$
- ☐ **E.**  $T(n)$  is  $\Theta(n)$

3. The function  $T(n)$  is assumed to have the initial value  $T(1) = 4$ . **Note:**  $\log n = \log(n) = \log_2(n)$ .

What is the correct solution to the following recurrence relation that holds for all  $n \geq 2$ :

$$T(n) = T(n-1) + 3.$$

- ☐ **A.**  $T(n) = 3n + 4$
- ☐ **B.**  $T(n) = 3n$
- ☒ **C.**  $T(n) = 3n + 1$
- ☐ **D.**  $T(n) = \Theta(n \log n)$
- ☐ **E.**  $T(n) = \Theta(n^2)$

4. Recall that Strassen's algorithm for multiplying two  $n \times n$  matrices follows a divide-and-conquer approach, and reduces the problem to 7 subproblems of size  $n/2 \times n/2$ . Which of the following is the correct worst-case running time of Strassen's algorithm described above obtained by the Master Method

- ☒ **A.**  $\Theta(n^{\log_2 7})$
- ☐ **B.**  $\Theta(n^2)$
- ☐ **C.**  $\Theta(n^3)$
- ☐ **D.**  $\Theta(n^{\log_3 7})$
- ☐ **E.**  $\Theta(n \log n)$

5. What is the expected running time of the randomised Select algorithm (where the pivot is chosen randomly at each step) to find the  $n/2$ 'th largest element in an array containing  $n$  elements?

- ☐ A.  $O(n \log n)$
- ☒ B.  $O(n)$
- ☐ C.  $O(\log n)$
- ☐ D.  $O(1)$
- ☐ E.  $O(n^2)$

6. What is the best estimate of the running time of Counting Sort on an array of  $n$  elements, each of which is an integer between 0 and  $k$ ?

- ☒ A.  $O(n + k)$
- ☐ B.  $O(n \log n)$
- ☐ C.  $O(n)$
- ☐ D.  $O(k)$
- ☐ E.  $O(n^2)$

7. What is the best estimate for the worst-case running time of HeapSort on an input array with  $n$  integers

- ☐ A.  $O(n)$
- ☐ B.  $O(n^2)$
- ☐ C.  $O(\log n)$
- ☐ D.  $O(1)$
- ☒ E.  $O(n \log n)$

8. Define the following sequence of numbers:  $T(1) = 3$ ,  $T(2) = 4$ , and  $T(n) = 2T(n-1) + T(n-2)$  for all integers  $n \geq 3$ . What is the value of  $T(6)$

- ☐ A. 11
- ☐ B. 34
- ☐ C. 111
- ☒ D. 152
- ☐ E. 2048

9. You are given an array of  $n$  numbers, each with  $d$  digits. Recall that the first step of RadixSort is to sort the least-significant digits of the inputs using a stable sorting algorithm. Suppose your input is

733  
273  
903  
915  
901

Which of the following is the correct order of elements after the first iteration

☐ **A.**

273  
733  
901  
903  
915

☐ **B.**

273  
733  
903  
915  
901

☒ **C.**

901  
733  
273  
903  
915

☐ **D.**

273  
733  
901  
915  
903

☐ **E.**

901  
273  
733  
903  
915

10. Consider the following instance to the **0-1 Knapsack problem**

| item    | 1  | 2  | 3  | 4  | 5 |
|---------|----|----|----|----|---|
| benefit | 42 | 21 | 22 | 36 | 9 |
| weight  | 6  | 3  | 4  | 5  | 2 |

Which of the following is the optimal benefit attainable if the knapsack has total capacity  $W = 10$

- ☐ A. 41
- ☒ B. 66
- ☐ C. 67
- ☐ D. 71
- ☐ E. 69

11. Consider the instance to the **Fractional Knapsack problem**

| item    | 1  | 2  | 3  | 4  | 5 |
|---------|----|----|----|----|---|
| benefit | 42 | 21 | 22 | 36 | 9 |
| weight  | 6  | 3  | 4  | 5  | 2 |

Which of the following is the optimal benefit attainable if the knapsack has total capacity  $W = 10$

- ☐ A. 41
- ☐ B. 66
- ☐ C. 67
- ☒ D. 71
- ☐ E. 69

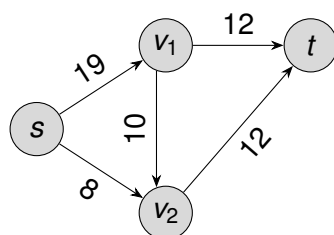
12. Let  $G$  be a simple undirected graph. Let  $n$  denote the number of vertices of  $G$  and let  $m$  denote the number of edges of  $G$ . Which of the following is always true when  $G$  is connected

- ☐ A.  $m = n$
- ☒ B.  $m \geq n - 1$
- ☐ C.  $m = 2n$
- ☐ D.  $m \leq n - 1$
- ☐ E.  $m \geq n$

13. Assuming edge relaxations take  $O(1)$  time, what is the best estimate of the worst-case running time of the Bellman-Ford algorithm on input graphs with  $n$  vertices and  $m$  edges

- ☐ A.  $O(m^2)$
- ☐ B.  $O(m + n)$
- ☐ C.  $O(m^n)$
- ☐ D.  $O(m)$
- ☒ E.  $O(mn)$

14. What is the value of the maximum flow from  $s$  to  $t$  in the following network flow



- ☐ A. 27
- ☐ B. 61
- ☐ C. 12
- ☐ D. 31
- ☒ E. 24

15. Alice wants to use a one-time pad to send a secure message to Bob. The message (plaintext) she wants to send is  $M = 1010$ , and the symmetric key  $K$  that they have agreed upon is  $K = 0011$ . Which of the following is the ciphertext

- ☐ A. 0110
- ☒ B. 1001
- ☐ C. 1100
- ☐ D. 0010
- ☐ E. 1010

16. Which of the following is true for all positive integers  $a, b$

- ☒ A.  $\gcd(a, b) = \gcd(b, a \bmod b)$
- ☐ B.  $\gcd(a, b) = \min(a, b)$
- ☐ C.  $\gcd(a, b) = \gcd(b, a/b)$
- ☐ D.  $\gcd(a, b) = a/b$
- ☐ E.  $\gcd(a, b) = a \bmod b$

17. Recall that in the RSA encryption scheme we choose  $n = p \cdot q$ , define  $\phi(n) = (p - 1)(q - 1)$ , and choose  $e, d$  such that  $\gcd(e, \phi(n)) = 1$  and  $ed \equiv 1 \pmod{\phi(n)}$ . Which of the following is the ciphertext corresponding to the plaintext  $M$ , obtained using the RSA scheme

- ☒ A.  $C = M^e \bmod n$
- ☐ B.  $C = M^d \bmod n$
- ☐ C.  $C = M$
- ☐ D.  $C = M^e \bmod \phi(n)$
- ☐ E.  $C = M^d \bmod \phi(n)$

18. Which of the following is *known to be true* about the complexity classes **P** and **NP**

☐ A.  $P \neq NP$

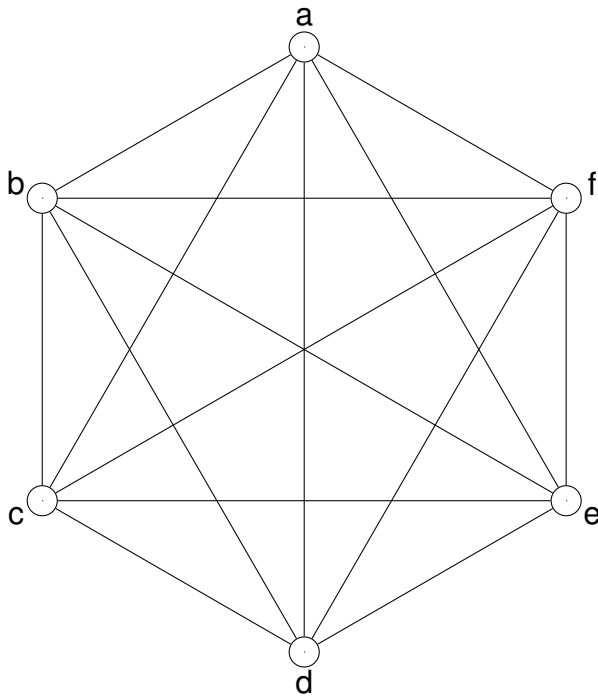
☐ B.  $P \supseteq NP$

☒ C. If there is a polynomial-time algorithm for 3SAT, then  $P = NP$

☐ D. All problems in **NP** have polynomial-time algorithms.

☐ E.  $P = NP$

19. Consider the following graph  $G$ , which is the complete graph on 6 vertices.



Which of the following is true about  $G$

☐ A. The smallest vertex cover of  $G$  has size 1

☐ B. The smallest vertex cover of  $G$  has size 2

☐ C. The smallest vertex cover of  $G$  has size 3

☒ D. The smallest vertex cover of  $G$  has size 5

☐ E. The smallest vertex cover of  $G$  has size 6

20. Recall that Fibonacci numbers are defined as  $F(1) = F(2) = 1$ , and  $F(n) = F(n-1) + F(n-2)$ . What is the best estimate for the number of bits required to represent  $F(n)$  in binary?

- ☒ A.  $O(n)$
- ☐ B.  $O(n^2)$
- ☐ C.  $O(1)$
- ☐ D.  $O(\log n)$
- ☐ E.  $O(\log^2 n)$